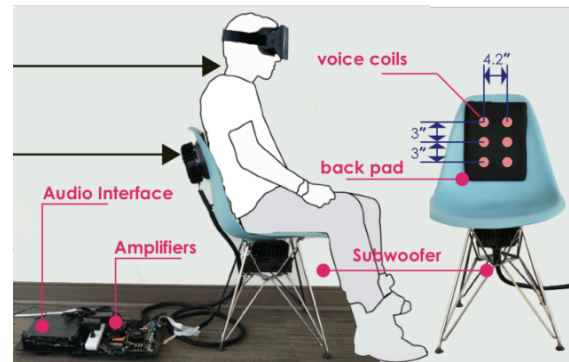


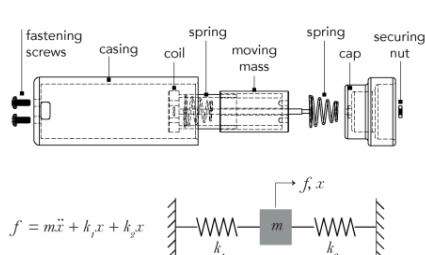
Augmented and Virtual haptics

VR360HD: We present a VR360° video player with haptic feedback playback. The VR360HD application enhances VR viewing experience by triggering customized haptic effects associated with user's activities, biofeedback, network messages and customizable timeline triggers incorporated in the VR media. The app is developed in the Unity3D game engine and tested using a GearVR headset, therefore allowing users to add animations to VR gameplay and to the VR360° streams. A custom haptic plugin allows users to author and associate animated haptic effects to the triggers, and playback these effects on a custom haptic hardware



- Israr, Ali, et al. "VR360HD: a VR360 player with enhanced haptic feedback." *Proceedings of the 22nd ACM Conference on Virtual Reality Software and Technology*. 2016. [\[Link\]](#)

Haplug (*hap-lug*) for dynamic VR interactions: The haplug is a linear voice coil motor packaged between springs in a plug-like casing. Response and bandwidth of the haplug are optimized for haptic interactions, and tuned by the motor inertia and spring stiffness. A pair of haplugs is mounted on the hand-tracking controllers of HTC Vive and renders a variety of haptic feedback, such as, feelings to colors, interactions with objects, surface texture and dynamic object behaviors. The Haptic Plug allows rich control of haptic effects in VR and other interactive settings.



- Hanamitsu, N., & Israr, A. (2016, November). Haplug: A haptic plug for dynamic VR interactions. In *International AsiaHaptics conference* (pp. 479-483). Singapore: Springer Singapore. [\[Link\]](#)
[@https://www.youtube.com/watch?v=ohhQ53A9fD8](https://www.youtube.com/watch?v=ohhQ53A9fD8)

Magic Bench is a multi-user & multi-sensory AR/MR platform, where a group can share the same augmented environment with any computer generated (CG) asset and interact in a shared story sequence through a third-person POV. Our approach is to instrument the environment leaving the user unburdened of any equipment, creating a seamless walk-up-and-play

experience. We demonstrate this technology in a series of vignettes featuring humanoid animals. Participants can not only see and hear these characters, they can also feel them on the bench through haptic feedback. Many of the characters also interact with users directly, either through speech or touch.

- McIntosh, K., Mars, J., Krahe, J., McCann, J., Rivera, A., Marsico, J., ... & Mahler, M. (2017). Magic bench: a multi-user & multi-sensory AR/MR platform. In *ACM SIGGRAPH 2017 VR Village* (pp. 1-2).

[\[Link\]](#)

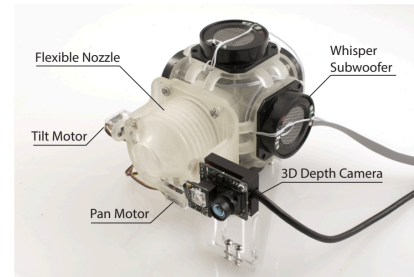
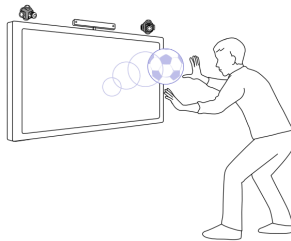
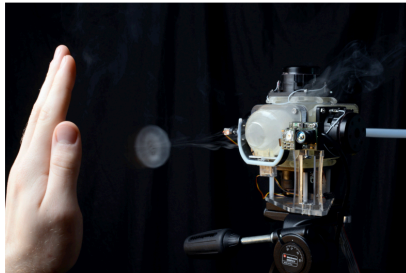


@  Magic Bench

AIREAL is a novel haptic technology that delivers effective and expressive tactile sensations in free air, without requiring the user to wear a physical device. Combined with interactive computers graphics, AIREAL enables users to feel virtual 3D objects, experience free air textures and receive haptic feedback on gestures performed in free space. AIREAL relies on air

vortex generation directed by an actuated flexible nozzle to provide effective tactile feedback with a 75 degrees field of view, and within an 8.5cm resolution at 1 meter.

- Sodhi, R., Poupyrev, I., Glisson, M., & Israr, A. (2013). AIREAL: interactive tactile experiences in free air. *ACM Transactions on Graphics (TOG)*, 32(4), 1-10. [[Link](#)]



@  AIREAL: Interactive Tactile Experiences in Free Air